

State and Trends in Salmon Abundance

(1) Metadata Reference Information

Metadata Last Updated: 20251112

Dataset URL: <https://data.salmonwatersheds.ca/result?datasetid=552>

Contact:

Organization: Pacific Salmon Foundation

Custodian: Stephanie Peacock

Position: Senior Analyst, Salmon Watersheds Program

Contact Address:

Address: #320 1385 W 8th Ave

City: Vancouver

Province: BC

Postal Code: V6H 3V9

Country: Canada

Contact Voice Telephone: 604 664 7664

Contact Email: speacock@psf.ca

(2) Identification Information

Data Identification:

Supplemental Information

-989898 is the PSF database code for a null value.

Attributes:

region	One of ten regions defined for organizing data in the PSE: Yukon, Transboundary, Haida Gwaii, Nass, Skeena, Central Coast, East Vancouver Island & Mainland Inlets, West Vancouver Island, Fraser, Columbia
species	One of six species: Chinook, chum, coho, pink, sockeye, steelhead
type	The type of data summarized: Spawners or Run Size (a.k.a. total abundance)
current_status	The percent difference (as a proportion 0 to 1) in the current abundance from the long-term average
short_trend	The average annual percent change in abundance based on the linear model fitted to the log-transformed, smoothed abundance over the most recent three generations, calculated as $\exp(\text{slope}) - 1$
short_trend_cat	Categorization of short_trend as decreasing (arrow-down), stable (arrow-left-right; slope not statistically significant at $p = 0.05$), or increasing (arrow-up).
long_trend	The average annual percent change in abundance based on the linear model fitted to the log-transformed, smoothed abundance over the entire time series, calculated as $\exp(\text{slope}) - 1$
long_trend_cat	Categorization of long_trend as decreasing (arrow-down), stable (arrow-left-right; slope not statistically significant at $p = 0.05$), or increasing (arrow-up).

current_abundance	The smoothed abundance in the most recent year, calculated as the geometric mean over the most recent generation (see sps-data.csv)
current_abundance_year	The most recent year of data for which current_abundance is available.
average_abundance	The geometric average abundance over the entire time series.
previous_gen_abundance	The smoothed abundance one generation previous to current abundance, calculated as the geometric mean over the second-most-recent generation (see sps-data.csv)
gen_length	The dominant generation length for each region and species, based on the reported generation length for component Conservation Units.
nyears	The number of years of abundance data for the region, species, and data type.
rangeyears	The range of years of abundance data for the region, species, and data type.
datasetversion	Date the dataset was last updated in PSF's database.

Extent

Description_of_Geographic_Extent: British Columbia and Yukon

Temporal extent:

Beginning_Date: 1893-01-01

Ending_Date: 2024-12-31

Status: Complete.

Maintenance and Update Frequency: As needed.

Citation:

Title: State and Trends in Salmon Abundance

Publication Date: 20251112

Responsible Party: Pacific Salmon Foundation

Publication Place: Vancouver, British Columbia

Contact Address:

Address: #320 1385 W 8th Ave

City: Vancouver

Province: BC

Postal Code: V6H 3V9

Country: Canada

Contact Voice Telephone: 604-664-7664

Contact Email: data@salmonwatersheds.ca

Description:

Abstract: This dataset includes metrics summarizing the time series of spawner and total abundance for each salmon species in nine regions.

Purpose: This dataset was developed by the Pacific Salmon Foundation's Salmon Watersheds Program as part of PSF's State of Salmon Report. The State of Salmon Report is a broad-scale evaluation of the state of all six species of Pacific salmon found in British Columbia and the Yukon, Canada. This flagship publication from the

Pacific Salmon Foundation takes a data-driven approach to summarizing the status and trends in regional abundance for each species of salmon in ten major salmon-bearing regions in western Canada.

Resource Constraints:

Use Limitations: There are no restrictions on the use of this data. However, secondary distribution must be accompanied by this documentation. Credit should always be given to the Pacific Salmon Foundation as the Originator and to the data sources when this data is transferred or printed.

Data Set Credit:

This dataset is based on core data from a variety of sources including the Pacific Salmon Commission, DFO, and LGL Ltd. Details of these sources are provided in the State of Salmon Report technical documentation available at <https://salmonwatersheds.github.io/state-of-salmon/>. The methods and models used to produce this dataset are based on published literature, agency reports, and knowledge of experts working with the species.

(3) Distribution Information

Distributor: Pacific Salmon Foundation
Publication Place: Vancouver, British Columbia
Contact Address:
Address: #320 1385 W 8th Ave
City: Vancouver
Province: BC
Postal Code: V6H 3V9
Country: Canada
Contact Voice Telephone: 604-664-7664
Contact Email: data@salmonwatersheds.ca

(4) Data Quality Information

Data Quality Constraints:

Data quality will vary depending on the source of the raw information.

Data Processing:

Data processing is detailed in the following GitHub repository: <https://github.com/salmonwatersheds/state-of-salmon>. Detailed description of the analytical approach are at: <https://salmonwatersheds.github.io/state-of-salmon/>.

Process Step 1:

Description: Compiled regional data for smoothed spawner and total abundance are read in and metrics calculated. Code file 3_calc-metrics.R (https://github.com/salmonwatersheds/state-of-salmon/blob/main/code/3_calc-metrics.R).

Date/time: 2024-07-19

Process Contact:

Contact Person: Stephanie Peacock

Contact Organization: Pacific Salmon Foundation (PSF)

Contact Role: Custodian

Country: Canada

Contact Email: speacock@psf.ca